SUBMISSION Documentation

V-SCANNER | MOBILE SECURITY SUITE

V-Scanner — Mobile App Vulnerability Scanner

A cross-platform security toolkit (Android App + CLI) that scans installed apps for vulnerabilities, monitors real-time sensor access, and generates actionable security reports.

Platform: Android | CLI

Language: Kotlin | Python

Android SDK: 26-35

License: MIT

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1. Problem Understanding

Modern Android devices contain **200+ installed apps** on average, each requesting a varying set of permissions. Users grant these permissions without fully understanding the privacy implications. The core problems are:

- **Permission Overreach** — Many apps request permissions far beyond their functional needs (e.g., a flashlight app requesting SMS and contacts access).

- **Outdated SDKs** — Apps targeting older Android SDKs bypass critical security features like runtime permissions (pre-Android 6.0) and background access restrictions.
- **Insecure Hardcoded Data** — Some APKs contain hardcoded HTTP URLs, API keys, Firebase credentials, and AWS secrets — making them vulnerable to interception and credential theft.
- **Silent Sensor Access** — Apps can access the camera, microphone, and GPS in the background without the user's knowledge, creating serious privacy violations.
- **No Centralized Visibility** — Android provides no single view for users to understand the combined security posture of all their installed apps.

V-Scanner addresses all five problems through a dual-platform approach: a native Android app for on-device scanning and real-time monitoring, and a Python CLI tool for remote forensic analysis via ADB.

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2. Key Challenges

Challenge	Description	Our Approach
False Positives	Legitimate apps (WhatsApp, Google Maps) naturally require many permissions. A naive scoring system would flag them as high-risk.	**Trust-based risk normalization** — Apps verified from Play Store get a 35% risk reduction; known trusted packages get an additional 30% reduction; system apps get 50% reduction.
Sensor Access Detection	`AppOpsManager.checkOpNoThrow()` only checks if a permission *policy* allows access —it doesn't detect *actual* sensor usage.	**Usage-time correlation** — We cross-reference `UsageStatsManager` events with permission grants and foreground/background state to identify actual sensor access patterns.
Background Monitoring Battery Impact	Continuously polling sensors drains the battery.	**Optimized 5-second polling intervals** with `PowerManager` integration — monitoring automatically adjusts based on screen state. The service runs as a lightweight foreground service with minimal wake locks.
API Compatibility (SDK 26–35)	Android APIs change across 10 major versions. `getInstallerPackageName()` is deprecated; `getInstallSourceInfo()` is API 30+.	**Dual-path API calls** with version checks — we use the modern API when available and gracefully degrade on older devices.
Permission Complexity	Android has 200+ permissions; categorizing and risk-weighting them requires domain expertise.	**Curated database of 28 dangerous permissions** across 11 categories, each with expert-assigned risk levels, human-readable descriptions, threat explanations, and secure alternative recommendations.
Sideloaded App Detection	Sideloaded apps bypass Play Store's safety checks and are inherently riskier.	**Multi-store installer verification** — We query the installer source (Play Store, Samsung Galaxy Store, Amazon Appstore, Huawei AppGallery) and flag apps from unknown sources.

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3. Proposed Solution & Technical Approach

V-Scanner is a **complete mobile security assessment platform** with two execution modes:

3.1 Native Android App (On-Device)

A Material3 Android application that runs directly on the device and provides:

- **One-tap full device scan** — Scans all installed apps, analyzes permissions, calculates risk scores, and generates a sortable/filterable security report.
- **Privacy Guardian** — A persistent foreground service that monitors camera, microphone, and location access in real-time, detects suspicious patterns (background access, screen-off access, excessive frequency), and pushes instant notifications.
- **Security Dashboard** — A at-a-glance view of the device's overall security posture with risk distribution, active alerts, and sensor usage trends.
- **Actionable Intelligence** — Each flagged permission includes a human-readable explanation of why it's dangerous, what the real-world risk is, and what the secure alternative is.

3.2 Python CLI Tool (Remote via ADB)

A cross-platform terminal tool for security researchers and power users:

- **Remote app scanning** — List, analyze, and report on apps installed on a connected Android device over ADB.
- **Live sensor monitoring** — Continuously poll `dumpsys camera|audio|location` to detect real-time hardware access.
- **APK binary analysis** — Grep APK files for insecure URLs, hardcoded credentials, API keys, Firebase config, and AWS secrets.
- **Report generation** — Export findings as interactive HTML reports, structured JSON, or formatted terminal output.
- **Device administration** — Uninstall apps, force-stop processes, gather full device info, and mirror the screen via `scrcpy`.

3.3 Why Two Platforms?

Aspect	Android App	CLI Tool
User	Everyday users	Security researchers
Access	On-device, no setup	Remote via ADB
Real-time	Yes (foreground service)	Yes (polling `dumpsys`)
Reporting	In-app UI	HTML / JSON / Text
Sensor Monitoring	Background Guardian	Live terminal feed
Device Admin	No	Yes (uninstall, force-stop)

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4. System Architecture

```
┌──────────────────────────────────────────────────────────────────────┐
│                          V-SCANNER PLATFORM                          │
├──────────────────────────────┬───────────────────────────────────────┤
│       ANDROID APP            │              CLI TOOL                 │
```

```
│   (On-Device Scanning)      │     (Remote via ADB)                │
│                             │                                     │
│   ┌─────────────────────┐   │   ┌──────────────────────────┐    │
│   │    Jetpack Compose  │   │   │     Rich Terminal UI     │    │
│   │  Material3 Dark UI  │   │   │   GEMINI-Inspired Design │    │
│   └──────────┬──────────┘   │   └──────────┬───────────────┘    │
│              │              │              │                    │
│   ┌──────────▼──────────┐   │   ┌──────────▼───────────────┐    │
│   │   MVVM ViewModels   │   │   │   Click CLI Framework    │    │
│   │  (StateFlow / Hilt) │   │   │   Interactive Menu System│    │
│   └──────────┬──────────┘   │   └──────────┬───────────────┘    │
│              │              │              │                    │
│   ┌──────────▼──────────┐   │   ┌──────────▼───────────────┐    │
│   │  AppScanner Engine  │   │   │   ADBInterface Scanner   │    │
│   │ PermissionDatabase  │   │   │   permissions.py Analyzer│    │
│   │ Risk Scoring Engine │   │   │   APK Binary Scanner     │    │
│   └──────────┬──────────┘   │   └──────────┬───────────────┘    │
│              │              │              │                    │
│   ┌──────────▼──────────┐   │   ┌──────────▼───────────────┐    │
│   │ Privacy Guardian    │   │   │   dumpsys Sensor Monitor │    │
│   │ Service (Foreground)│   │   │   Live Camera/Mic/GPS Feed│    │
│   │ 5-sec Polling Loop  │   │   │   2-sec Polling Loop     │    │
│   └──────────┬──────────┘   │   └──────────┬───────────────┘    │
│              │              │              │                    │
│   ┌──────────▼──────────┐   │   ┌──────────▼───────────────┐    │
│   │  Room Database v2   │   │   │   Jinja2 HTML Reports    │    │
│   │  DataStore Prefs    │   │   │   JSON / Text Export     │    │
│   │  5 Tables, 5 DAOs   │   │   │   Console Rich Tables    │    │
│   └─────────────────────┘   │   └──────────────────────────┘    │
│                             │                                     │
│  ◆ PackageManager API       │  ◆ ADB Shell Commands               │
│  ◆ AppOpsManager API        │  ◆ dumpsys (camera, audio, GPS)     │
│  ◆ UsageStatsManager API    │  ◆ pm list packages                 │
│  ◆ PowerManager API         │  ◆ APK binary grep                  │
└─────────────────────────────┴─────────────────────────────────────┘
```

Android App — Internal Architecture (Clean MVVM)

```
┌──────────────────────────────────────────────────────┐
│                   UI Layer                           │
│   DashboardScreen  ScannerScreen  GuardianScreen     │
│   AlertsScreen     SettingsScreen                    │
└──────────────────────┬───────────────────────────────┘
                       │ Observes StateFlow
┌──────────────────────▼───────────────────────────────┐
│              ViewModel Layer                         │
│  DashboardVM  ScannerVM  GuardianVM  AlertsVM        │
│  SettingsVM    (@HiltViewModel / Inject)             │
```

```
└──────────────────────────┬──────────────────────────────┘
                           │
                           │ Calls suspend functions
┌──────────────────────────▼──────────────────────────────┐
│            Repository Layer                              │
│    ScannerRepository    GuardianRepository               │
└──────────────────────────┬──────────────────────────────┘
                           │
                           │ Room DAO queries
┌──────────────────────────▼──────────────────────────────┐
│              Data Layer                                  │
│   VScannerDatabase (Room v2) + DataStore Prefs           │
│   5 Entities │ 5 DAOs │ 16 Preference Keys               │
└──────────────────────────────────────────────────────────┘
```

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5. Technology Stack

Android Application

Technology	Version	Purpose
Kotlin	1.9.20	Primary language with coroutines & Flow
Jetpack Compose	BOM 2023.10.01	Declarative UI framework
Material3	BOM-managed	Design system with dynamic colors
Dagger Hilt	2.48	Compile-time dependency injection
Room	2.6.1	Local SQLite ORM with Flow-based queries
Jetpack DataStore	1.0.0	Type-safe preferences storage
Navigation Compose	2.7.5	Screen routing & bottom navigation
WorkManager	2.9.0	Reliable background task scheduling
Vico Charts	1.13.0	7-day usage trend visualization
Coroutines	1.7.3	Structured concurrency
Gson	2.10.1	JSON serialization for Room converters
Android SDK	26–35	Supports Android 8.0 to Android 15

CLI Tool

Technology	Version	Purpose
Python	3.x	Runtime
Rich	≥13.0.0	Terminal tables, panels, progress bars
Click	≥8.0.0	CLI argument parsing
Jinja2	≥3.0.0	HTML report templating
adb-shell	≥0.4.3	Python ADB bridge
scrcpy	External	Real-time screen mirroring

Build & Infrastructure

Tool	Version
Gradle	8.2
Android Gradle Plugin	8.2.0
JDK	17 (Android Studio JBR)
ProGuard/R8	Enabled for release builds
Git	Version control

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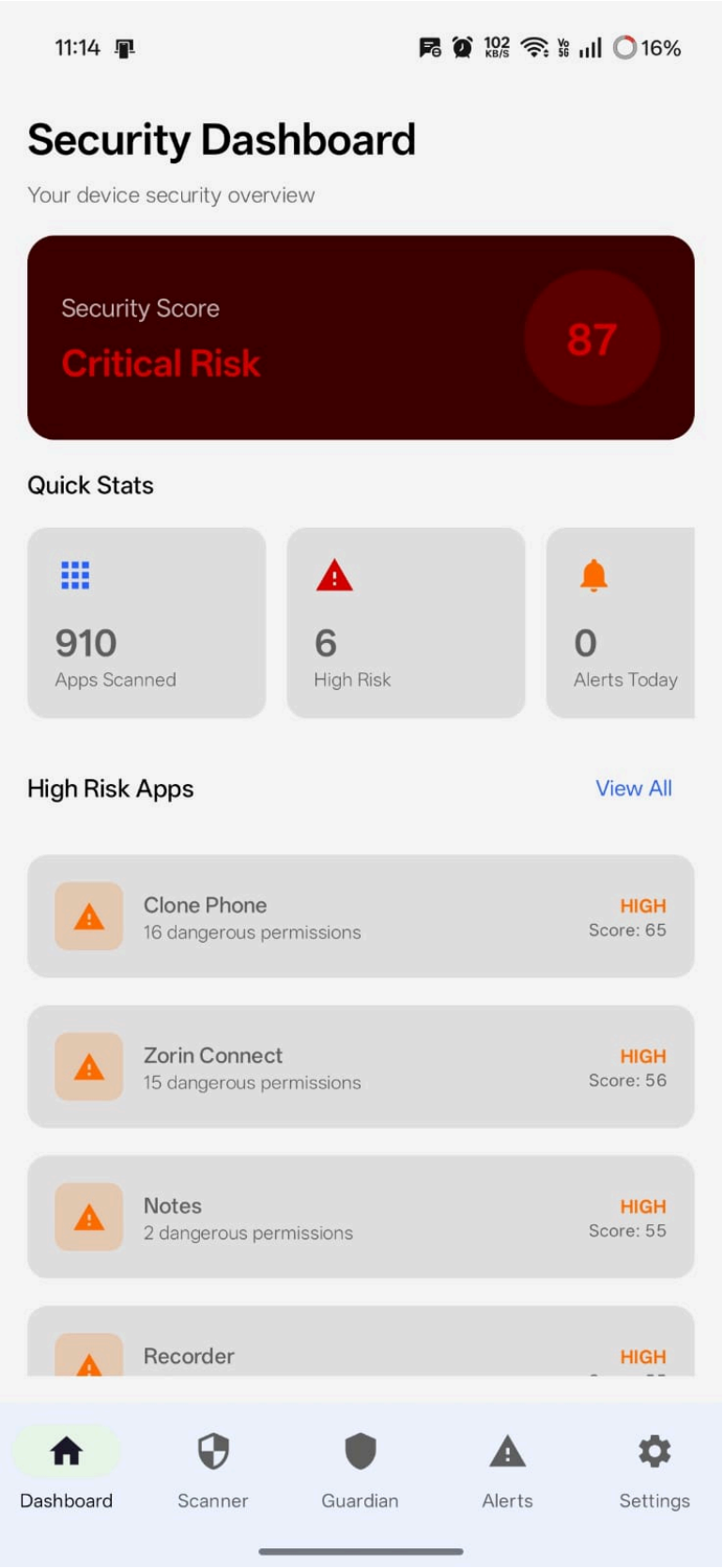
6. Feature Breakdown

6.1 Android App — 5 Core Screens

Dashboard

> The central security overview — at a glance, users understand their device's security posture.

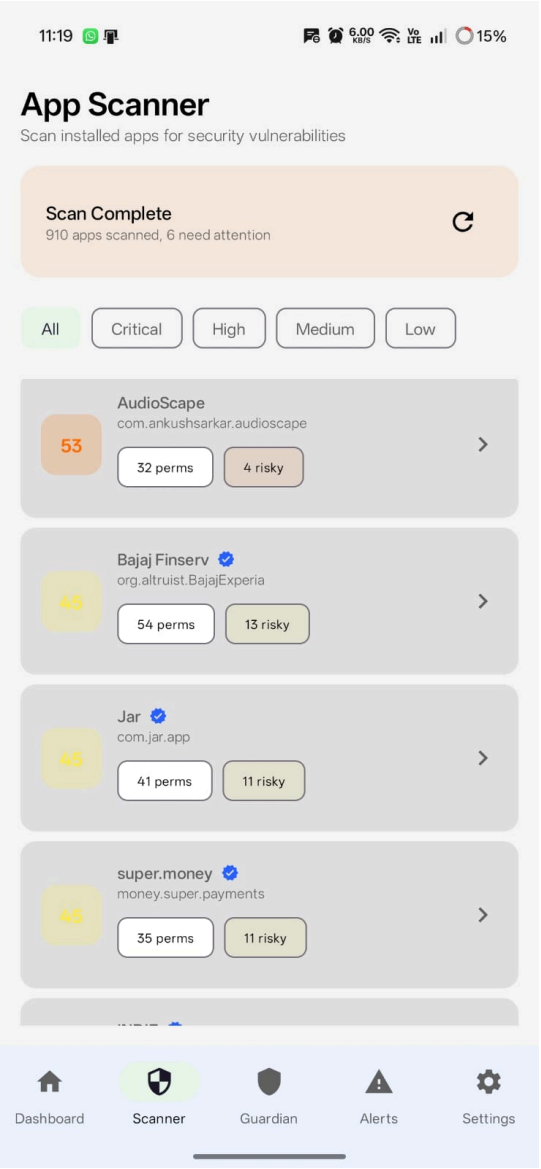
- **Overall Security Score** (0–100) with color-coded gauge
- **Quick Stats Strip**: Apps Scanned · High Risk Apps · Alerts Today · Guardian Status
- **Today's Sensor Usage**: Camera, Microphone, Location, Background access counts
- **Recent Alerts**: Last 3 privacy alerts with sensor icons
- **High Risk Apps**: Top 5 apps by risk score
- **Quick Actions**: Scan Now button, Guardian toggle

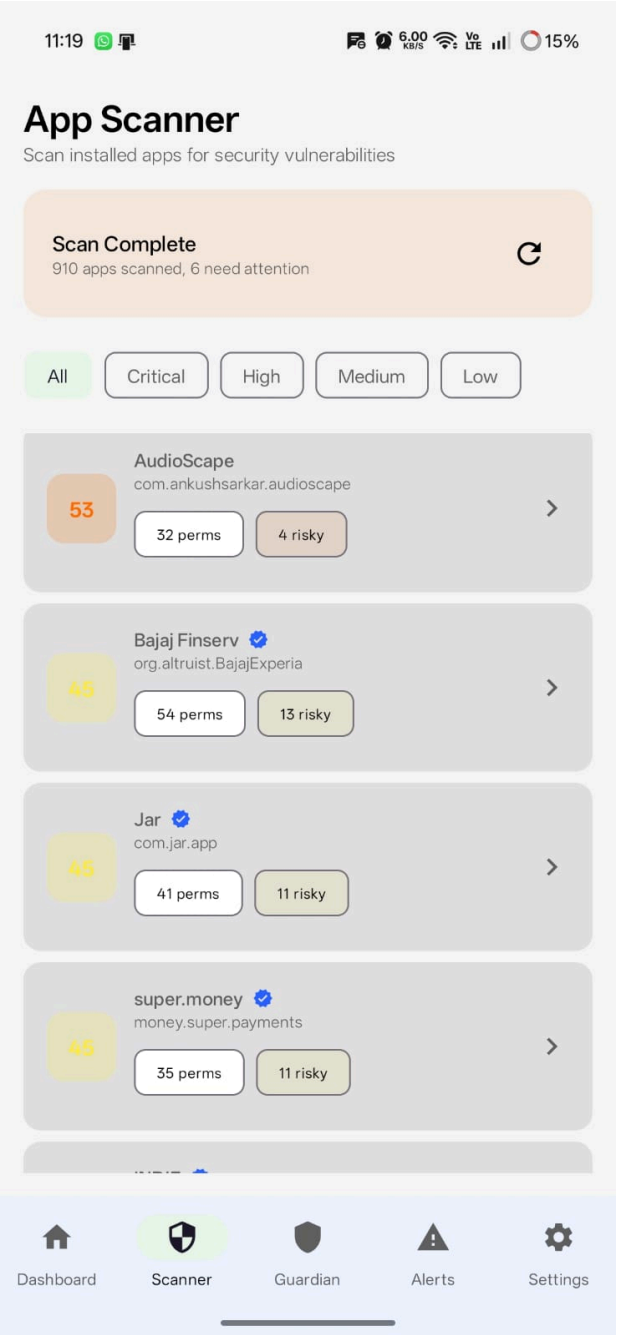

11:14
16%
Security Dashboard
Your device security overview
Security Score
Critical Risk
87
Quick Stats
910
Apps Scanned
6
High Risk
0
Alerts Today
High Risk Apps
View All
Clone Phone
16 dangerous permissions
HIGH
Score: 65
Zorin Connect
15 dangerous permissions
HIGH
Score: 56
Notes
2 dangerous permissions
HIGH
Score: 55
Recorder
HIGH
Dashboard
Scanner
Guardian
Alerts
Settings

Scanner

> Scans all installed apps and produces a detailed security assessment with color-coded risk scores.

- **One-tap scan** with real-time progress bar and per-app status
- **Risk-categorized app list** (sorted by score descending)
- **Filter chips**: All / Critical / High / Medium / Low
- **Per-app card** shows:
  - Color-coded risk score badge (0–100)
  - App name with ✅ **Play Store verified blue tick** for verified apps
  - Total permissions count + dangerous permissions count
- **Detailed bottom sheet** (tap any app):
  - Risk level breakdown with circular progress indicator
  - SDK information (Target SDK, Min SDK)
  - **Installer source** (Google Play Store / Samsung Galaxy Store / Unknown / Pre-installed)
  - Install date and last update date
  - Dangerous permissions grouped by category with risk badges
  - Per-permission: description, risks, and mitigation recommendations
  - Safe permissions summary count

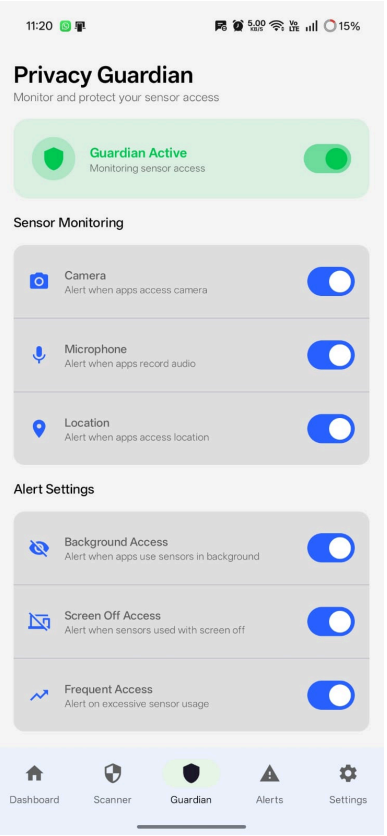

11:19
15%
App Scanner
Scan installed apps for security vulnerabilities
Scan Complete
910 apps scanned, 6 need attention
All
Critical
High
Medium
Low
53
AudioScape
com.ankushsarkar.audioscape
32 perms
4 risky
45
Bajaj Finserv
org.altruist.BajajExperia
54 perms
13 risky
45
Jar
com.jar.app
41 perms
11 risky
45
super.money
money.super.payments
35 perms
11 risky
Dashboard
Scanner
Guardian
Alerts
Settings



Privacy Guardian

> Real-time sensor monitoring service that detects and alerts on suspicious app behavior.

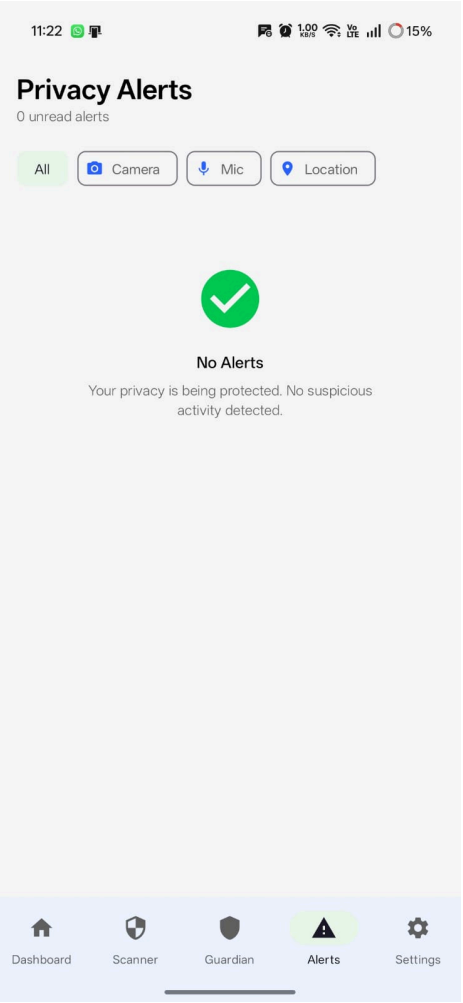
- **Master toggle** to enable/disable Guardian service
- **Per-sensor toggles**: Camera · Microphone · Location
- **Alert condition toggles**: Background Access · Screen-Off Access · Frequent Access
- **Recently Active Apps** section with per-sensor access counts
- **7-Day Sensor Usage Trend** chart (Vico bar chart)
- **Daily Summary** cards



Alerts

> Chronological list of all privacy alerts triggered by the Guardian service.

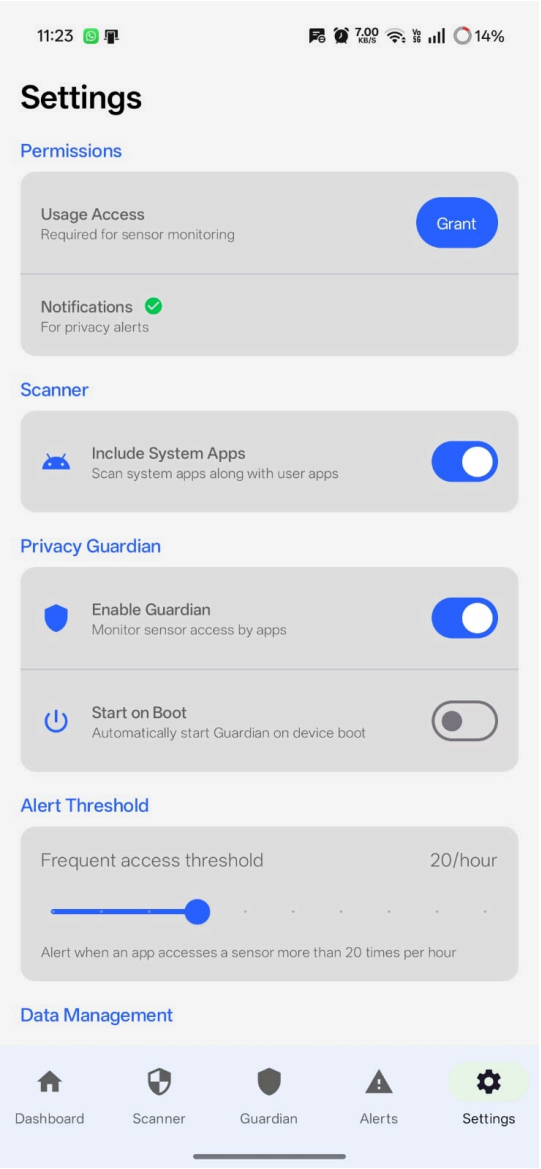
- **Unread count** badge in header
- **Sensor type filter chips**: Camera / Microphone / Location
- **Alert cards** with: sensor icon + color, app name, alert message, relative timestamp, alert type badge (Background / Frequent / Screen-Off / Suspicious)
- **Dismiss individual alerts** or **Clear All**
- **Empty state** with green checkmark when no alerts exist



Settings

> Full configuration control for scanner behavior, Guardian tuning, data management, and appearance.

- **Permissions Section**: Usage Access and Notification permission status with Grant buttons (status **updates live** when returning from system settings)
- **Scanner**: Include System Apps toggle
- **Privacy Guardian**: Enable Guardian, Start on Boot toggles
- **Alert Threshold**: Configurable slider (5–50 per hour)
- **Data Management**: Clear Scan History / Clear Sensor Logs / Clear All Alerts (with confirmation dialogs)
- **Appearance**: Dark / Light theme toggle
- **About**: Version info



6.2 CLI Tool — Key Features

Feature	Description
List All Apps	Enumerate installed packages with intelligent name resolution
Single App Analysis	Deep-dive: permissions, SDK risk, insecure URL detection
Full Device Scan	Batch analyze all apps with configurable depth
Live Sensor Monitor	Real-time camera/mic/GPS usage feed (2-sec refresh)
Hardware Sensor Values	CPU-Z style: accelerometer, gyroscope, magnetometer, proximity, light, barometer
Full Device Info	Hardware, System, Memory, Network, Identifiers, Build info panels
APK Binary Analysis	Detect insecure HTTP URLs, hardcoded API keys, Firebase config, AWS credentials
HTML Report Export	Dark-themed interactive HTML reports with expandable app cards and risk meters
JSON/Text Export	Machine-readable output for CI/CD integration
Screen Mirroring	Live screen share via `scrcpy` with mouse/keyboard
App Administration	Uninstall, force-stop, launch apps remotely

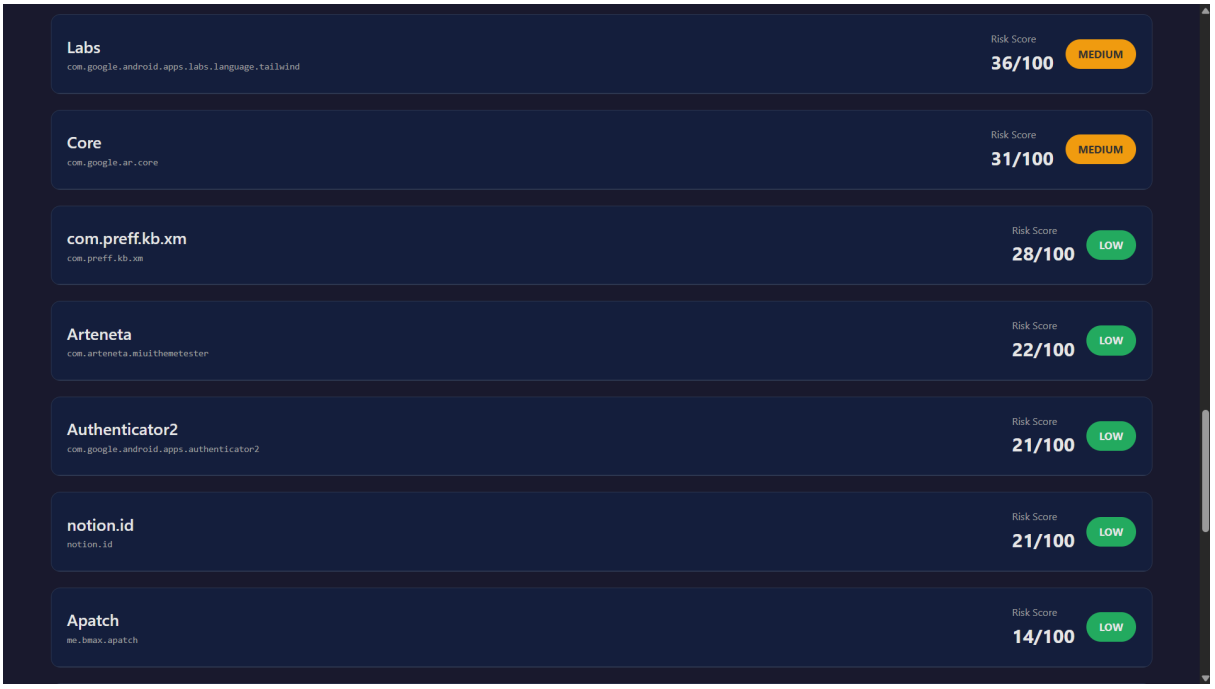
```
C:\Windows\System32\cmd.e

MOBILE    APPLICATION    SECURITY    SCANNER

🔐 Scan • Analyze • Protect • Defend

Version 2.0 | Enterprise Security | Real-time Analysis

... ✓ ▸ Initializing Security Engine
... ✓ ▸ Loading Vulnerability Database
... ✓ ▸ Connecting to Android Device
... ✓ ▸ Syncing Device Configuration
... ✓ ▸ Preparing Analysis Framework

⚙ Configuring Android Debug Bridge...
✓ Using saved ADB configuration
✓ ADB ready: D:\Gitam Leaker\V Scanner\platform-tools\adb.exe

🔍 Scanning for Android devices...

✓ Auto-selected device: 13d027a7

📱 Device: 13d027a7
```

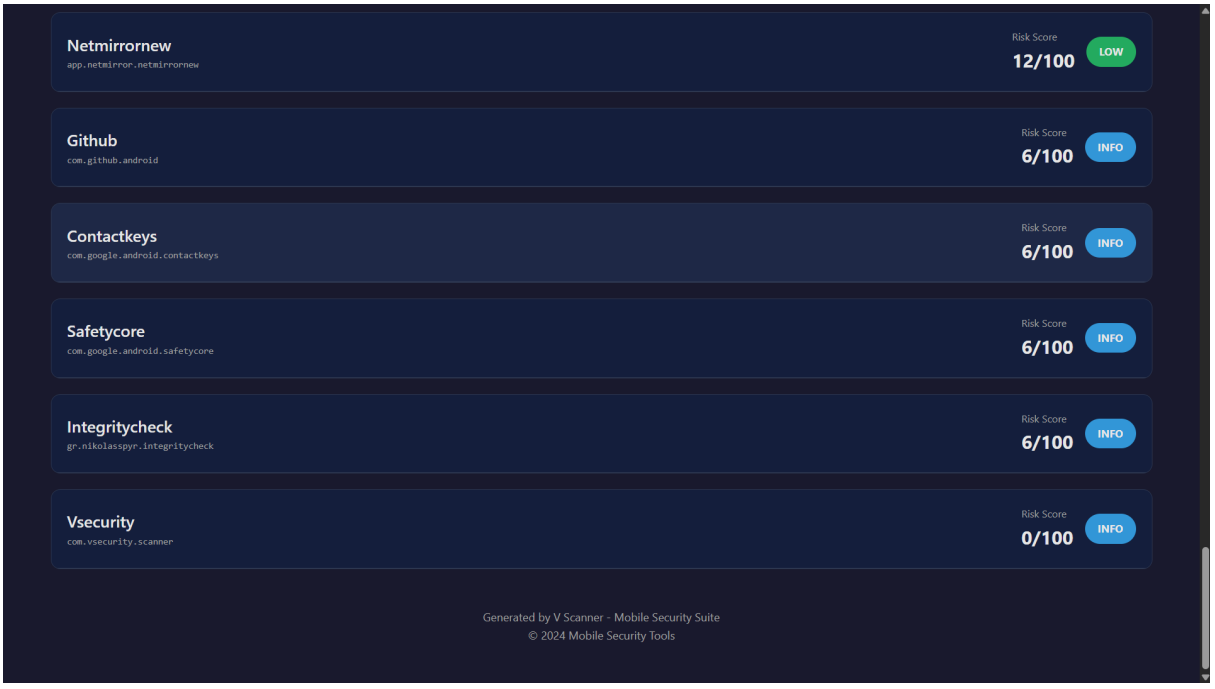
```
C:\Windows\System32\cmd.e

📊 Security Analysis: Messenger

Package:  org.telegram.messenger
Version: 12.4.3 (Code: 65272)
Target SDK: 35
Min SDK: 23

🔴 DANGEROUS PERMISSIONS (18):
  • Unknown - Allows approximate location access
  • Unknown - Allows precise GPS location access
  • Unknown - Allows initiating phone calls
  • Unknown - Allows reading call history
  • Unknown - Allows internet access
  • Unknown - Allows audio recording
  • Unknown - Allows viewing Wi-Fi connection info
  • Unknown - Allows reading external storage
  • Unknown - Allows reading contact data
  • Unknown - Allows modifying contact data
  • Unknown - Allows drawing over other apps
  • Unknown - Allows reading phone state and identity
  • Unknown - Allows access to device camera
  • Unknown - Allows requesting package installation
  • Unknown - Allows writing to external storage
  • Unknown - Allows location access in background
  • Unknown - Allows internet access
  • Unknown - Allows viewing Wi-Fi connection info

⚠ SDK Issues:
  • Min SDK 23 allows installation on insecure Android versions

⚠ RISK LEVEL: CRITICAL (Score: 100/100)

💡 Recommendations:
  • [ACCESS_COARSE_LOCATION] Prefer over fine location
  • [ACCESS_COARSE_LOCATION] Review necessity
  • [ACCESS_FINE_LOCATION] Use 'While using app' option
  • [ACCESS_FINE_LOCATION] Deny background location
  • [CALL_PHONE] Use dial intent instead
  • [CALL_PHONE] Review call logs
  • [READ_CALL_LOG] Deny to most apps
  • [READ_CALL_LOG] Clear call logs periodically
  • [INTERNET] Monitor data usage
  • [INTERNET] Use firewall apps
  • [RECORD_AUDIO] Grant only to call/voice apps
  • [RECORD_AUDIO] Revoke when not needed
  • [ACCESS_WIFI_STATE] Standard permission, low risk
  • [READ_EXTERNAL_STORAGE] Use scoped storage
  • [READ_EXTERNAL_STORAGE] Grant selective media access
  • [READ_CONTACTS] Grant only to trusted communication apps
  • [READ_CONTACTS] Review app reviews before granting
  • [WRITE_CONTACTS] Deny to most apps
  • [WRITE_CONTACTS] Backup contacts regularly
  • [SYSTEM_ALERT_WINDOW] Deny to untrusted apps
  • [SYSTEM_ALERT_WINDOW] Review overlay permissions in settings
  • [READ_PHONE_STATE] Deny if not call-related
  • [READ_PHONE_STATE] Review data collection policies
  • [CAMERA] Grant only when actively using camera features
  • [CAMERA] Revoke when not needed
  • [REQUEST_INSTALL_PACKAGES] Deny to most apps
  • [REQUEST_INSTALL_PACKAGES] Install only from trusted sources
  • [WRITE_EXTERNAL_STORAGE] Use app-specific directories
  • [WRITE_EXTERNAL_STORAGE] Review file access patterns
  • [ACCESS_BACKGROUND_LOCATION] Deny to most apps
  • [ACCESS_BACKGROUND_LOCATION] Allow only for navigation/fitness apps
  • [INTERNET] Monitor data usage
```

```
🔴 Live Hardware Usage Monitor
Real-time detection of active hardware access...
Last update: 23:23:15

⚠ Unable to determine current app

📷 Camera Status
  🔴 ACTIVE Provider (android.hardware.camera.provider)
  🔴 ACTIVE Camera (com.android.camera)

🎤 Microphone Status
  🔴 ACTIVE Recorderservice (df24232com.android.soundrecorder.recorderservice)
  🔴 ACTIVE Soundrecorder (com.android.soundrecorder)

📍 Location (GPS) Status
  🔴 ACTIVE Google Play Services (com.google.android.gms)
  🔴 ACTIVE Nianticlabs (com.nianticlabs.pokemongo)

📊 Summary
  Camera apps: 2 active
  Mic apps: 2 active
  GPS apps: 2 active

⚠ Apps are actively using hardware

[Press Ctrl+C to exit]

🔴 Live Hardware Usage Monitor
Real-time detection of active hardware access...
Last update: 23:23:19
```

> **[SCREENSHOT PLACEHOLDER — CLI HTML Report: Show the generated dark-themed HTML report with app cards and risk meters]**

Mobile Security Scan Report
Generated: 2026-03-03 23:03:54
Device: Redmi Note 9 Pro Max (Android 12)
33
TOTAL APPS SCANNED
19
HIGH RISK APPS
2
MEDIUM RISK APPS
12
LOW RISK APPS
Application Security Details
Telegram
org.telegram.messenger
Risk Score
100/100
CRITICAL
WhatsApp
com.whatsapp
Risk Score
100/100
CRITICAL
Instagram
com.instagram.android
Risk Score
100/100
CRITICAL
Videoplayer
Risk Score
CRITICAL


Labs
com.google.android.apps.labs.language.tailwind
Risk Score
36/100
MEDIUM
Core
com.google.ar.core
Risk Score
31/100
MEDIUM
com.preff.kb.xm
com.preff.kb.xm
Risk Score
28/100
LOW
Arteneta
com.arteneta.miuithemetester
Risk Score
22/100
LOW
Authenticator2
com.google.android.apps.authenticator2
Risk Score
21/100
LOW
notion.id
notion.id
Risk Score
21/100
LOW
Apatch
me.bmax.apatch
Risk Score
14/100
LOW



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7. Risk Scoring Algorithm — Deep Dive

Our risk scoring algorithm avoids the **false-positive trap** where legitimate apps like WhatsApp or Google Maps would score as "Critical" simply because they require many permissions.

Scoring Pipeline

```
Step 1: Permission Weight Accumulation
──────────────────────────────────────

For each dangerous permission found:
  CRITICAL permission → +12 points
  HIGH     permission → +7  points
  MEDIUM   permission → +3  points
  LOW      permission → +1  point
  INFO     permission → +0  points

Step 2: Normalization (Prevents Inflation)
──────────────────────────────────────────

  average_weight = raw_score / number_of_dangerous_permissions
  normalized_score = average_weight × 5     (max ~60 from averages)

Step 3: Critical Permission Bonus
─────────────────────────────────

  score += critical_permission_count × 8    (SMS, background location, ove
rlay)

Step 4: SDK Age Penalty
───────────────────────

  targetSdk < 23 (pre-Marshmallow) → +15   (no runtime permissions!)
```

```
  targetSdk < 26 (pre-Oreo)        → +8
  targetSdk < 28 (pre-Pie)         → +3

Step 5: Trust Reduction ★ (Unique to V-Scanner)
──────────────────────────────────────────────

  Play Store verified   → score × 0.65  (35% reduction)
  Known trusted package → score × 0.70  (30% reduction)
  System app            → score × 0.50  (50% reduction)

Step 6: Classification
──────────────────────

  ≥ 70 → CRITICAL (Red)
  ≥ 50 → HIGH     (Orange)
  ≥ 25 → MEDIUM   (Yellow)
  ≥ 8  → LOW      (Green)
  < 8  → INFO     (Blue)
```

Why This Works

Scenario	Naive Approach	V-Scanner
WhatsApp (Camera + Mic + Location + Contacts + Storage)	Score: 53 → HIGH ❌	Score: 15 → LOW ✅ (Play Store + trusted package reductions)
Random sideloaded app with SMS + Overlay + Background Location	Score: 75 → CRITICAL	Score: 82 → CRITICAL ✅ (no trust reduction, CRITICAL bonuses)
Google Maps (Fine Location + Coarse Location + Camera)	Score: 30 → MEDIUM ❌	Score: 8 → LOW ✅ (Play Store + Google package reductions)
Sideloaded flashlight with SMS + Contacts + Phone	Score: 48 → MEDIUM	Score: 62 → HIGH ✅ (no trust, suspicious permission combo)

Multi-Store Installer Detection

V-Scanner doesn't just check Google Play — it identifies installers from **5 app stores**:

Installer Package	Store Name
`com.android.vending`	Google Play Store
`com.sec.android.app.samsungapps`	Samsung Galaxy Store
`com.amazon.venezia`	Amazon Appstore
`com.huawei.appmarket`	Huawei AppGallery
`null` (system app)	Pre-installed
Any other / `null`	Unknown (flagged)

8. Privacy Guardian — Real-Time Sensor Monitoring

How It Works

```
┌──────────────────────────────────────────────────┐
│          PrivacyGuardianService                  │
```

```
│         (Foreground Service)                    │
│                                                 │
│   ┌──────────────────────────────────────────┐  │
│   │  Every 5 seconds:                        │  │
│   │                                          │  │
│   │  1. Check: Is screen on or off?          │  │
│   │     └─ PowerManager.isInteractive        │  │
│   │                                          │  │
│   │  2. Identify foreground app              │  │
│   │     └─ UsageStatsManager.queryEvents     │  │
│   │                                          │  │
│   │  3. For each sensor (Camera/Mic/GPS):    │  │
│   │     └─ Is monitoring enabled? (prefs)    │  │
│   │     └─ AppOpsManager: who has access?    │  │
│   │     └─ Cross-ref with recent activity    │  │
│   │     └─ Is app in background?             │  │
│   │     └─ Is screen off?                    │  │
│   │     └─ Is access frequency suspicious?   │  │
│   │                                          │  │
│   │  4. If suspicious → Log + Alert          │  │
│   │     └─ Insert SensorAccessLog (Room)     │  │
│   │     └─ Insert PrivacyAlert (Room)        │  │
│   │     └─ Push notification (high priority)│  │
│   │     └─ Update DailySensorSummary         │  │
│   └──────────────────────────────────────────┘  │
│                                                 │
│  Settings synced live from DataStore:           │
│  • monitorCamera / monitorMicrophone            │
│  • monitorLocation / alertOnBackground          │
│  • alertOnScreenOff / alertOnFrequent           │
│  • frequentThreshold / whitelistedApps          │
└─────────────────────────────────────────────────┘
```

Alert Types

Alert Type	Trigger Condition	Severity
`BACKGROUND_SENSOR_ACCESS`	App accesses sensor while not in foreground	High
`SCREEN_OFF_ACCESS`	App accesses sensor while screen is off	Critical
`FREQUENT_ACCESS`	App exceeds threshold (default: 10/hour) per sensor	Medium
`SUSPICIOUS_PATTERN`	Combination of background + screen-off + frequent	Critical

Notifications

Each alert pushes a high-priority notification with:
- Sensor-specific emoji (📷 Camera / 🎤 Microphone / 📍 Location)
- App name and alert description
- Tap-to-open action launching the Alerts screen

Boot Persistence

The `BootReceiver` listens for both `BOOT_COMPLETED` and `QUICKBOOT_POWERON` intents, reads the Guardian preference, and auto-starts the service if it was previously enabled.

---

9. Permission Intelligence Database

V-Scanner maintains a curated database of **28 dangerous Android permissions** across **11 categories**.

Each entry contains:
- **Risk Level** (CRITICAL / HIGH / MEDIUM / LOW / INFO)
- **Human-Readable Name** (e.g., "Read SMS" instead of `android.permission.READ_SMS`)
- **Plain-English Description** of what the permission allows
- **Real-World Risks** — What can go wrong (e.g., "Can read OTP/2FA codes", "Unauthorized SMS charges")
- **Recommended Mitigations** — What the user should do
- **Secure Alternatives** — What the developer should have used instead

Permission Categories

Category	Permissions Tracked	Highest Risk
SMS	READ_SMS, SEND_SMS, RECEIVE_SMS	🔴 CRITICAL
Location	FINE_LOCATION, COARSE_LOCATION, BACKGROUND_LOCATION	🔴 CRITICAL
Storage	READ_STORAGE, WRITE_STORAGE, MANAGE_ALL_FILES	🔴 CRITICAL
Special	SYSTEM_ALERT_WINDOW, REQUEST_INSTALL_PACKAGES	🔴 CRITICAL
Camera	CAMERA	🟠 HIGH
Microphone	RECORD_AUDIO	🟠 HIGH
Contacts	READ_CONTACTS, WRITE_CONTACTS, GET_ACCOUNTS	🟠 HIGH
Phone	READ_PHONE_STATE, CALL_PHONE, READ/WRITE_CALL_LOG, PROCESS_OUTGOING_CALLS	🟠 HIGH
Calendar	READ_CALENDAR, WRITE_CALENDAR	🟡 MEDIUM
Sensors	BODY_SENSORS, ACTIVITY_RECOGNITION	🟡 MEDIUM
Network	INTERNET, ACCESS_WIFI_STATE	🟢 LOW

---

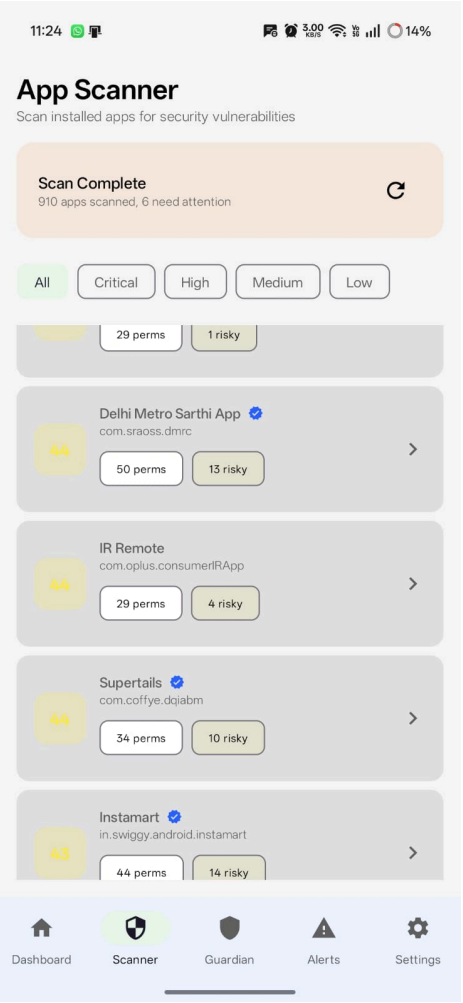
10. Innovative Features & Unique Angles

✅ Play Store Verified Badge (Blue Tick)

This is a standout feature. In the Scanner screen, every app installed from the Google Play Store displays a **verified blue checkmark** (✅) next to its name — similar to social media verification badges.

This instantly communicates trust to users:
- **Blue tick present** → App came from a verified store → Lower risk
- **No blue tick** → App was sideloaded or from an unknown source → Higher scrutiny needed

The verification extends beyond just Google Play — we detect **5 different app stores** and display the installer source in the app detail sheet.



Other Innovative Features

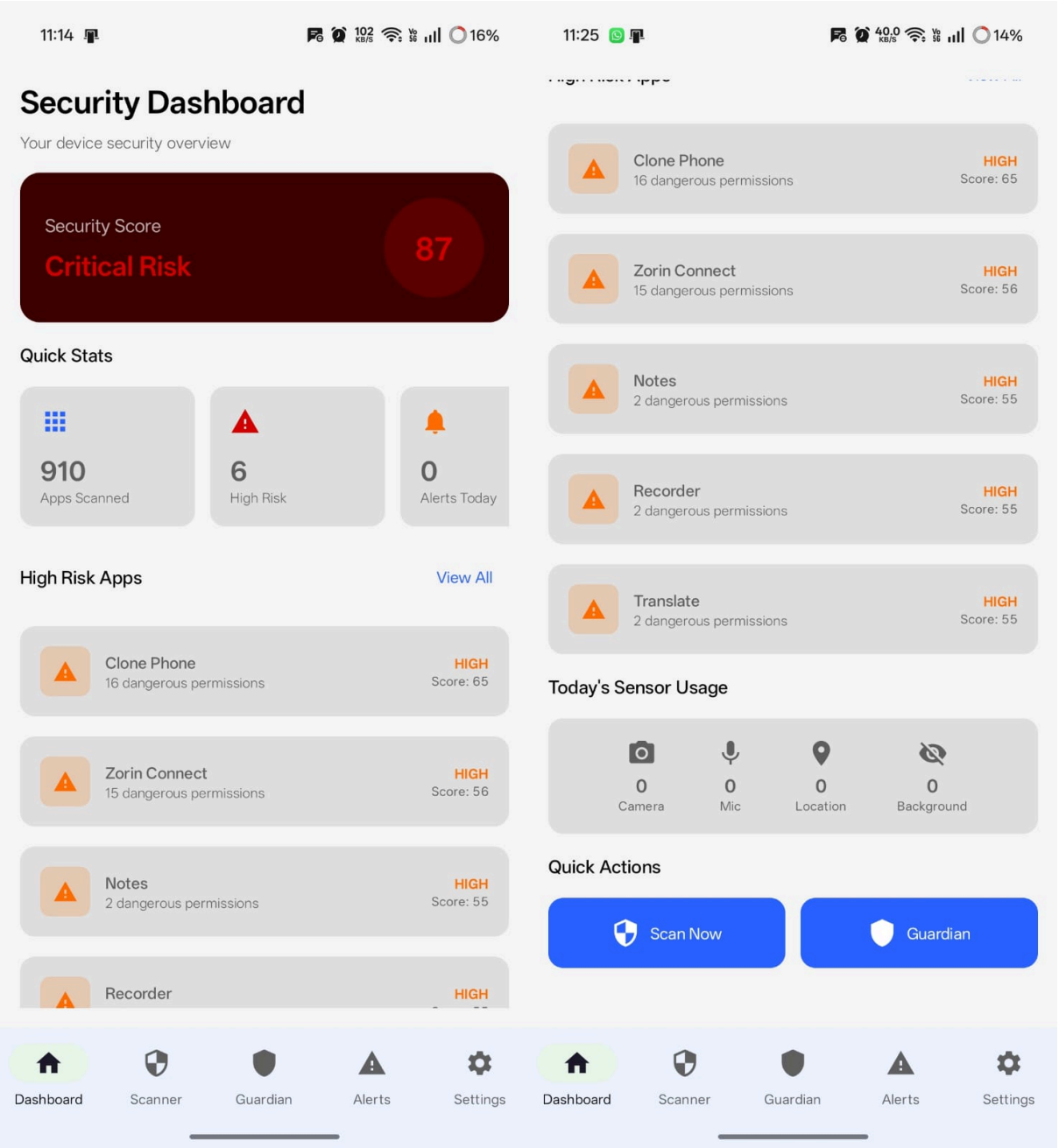
Feature	Innovation
Trust-Based Risk Normalization	Unlike static scoring, our algorithm considers the *source* of the app. A banking app from Play Store with many permissions scores differently than an unknown app with the same permissions.
Zero Network Architecture	V-Scanner makes **zero network calls**. All analysis, logging, and reporting happens 100% on-device. No data ever leaves the phone — true privacy-first design.
Screen-Off Sensor Detection	We flag apps accessing camera/microphone/GPS when the screen is off — a strong indicator of surveillance or spyware behavior.
Frequency-Based Anomaly Detection	Configurable threshold (5–50/hour) detects apps that access sensors obsessively, even if each individual access seems benign.
Boot-Persistent Guardian	Guardian service survives device reboots via `BOOT_COMPLETED` receiver — continuous protection without user intervention.
Cross-Platform Parity	Both CLI and Android app perform the same core analysis, ensuring consistent results regardless of the tool used.
APK Binary Scanning (CLI)	The CLI tool doesn't just check permissions — it scans APK binaries for hardcoded secrets: HTTP URLs, API keys, Firebase credentials, AWS access keys.
Live Sensor Feed (CLI)	Real-time 2-second polling of `dumpsys camera`, `dumpsys audio`, and `dumpsys location` with process identification — shows exactly which app is using which

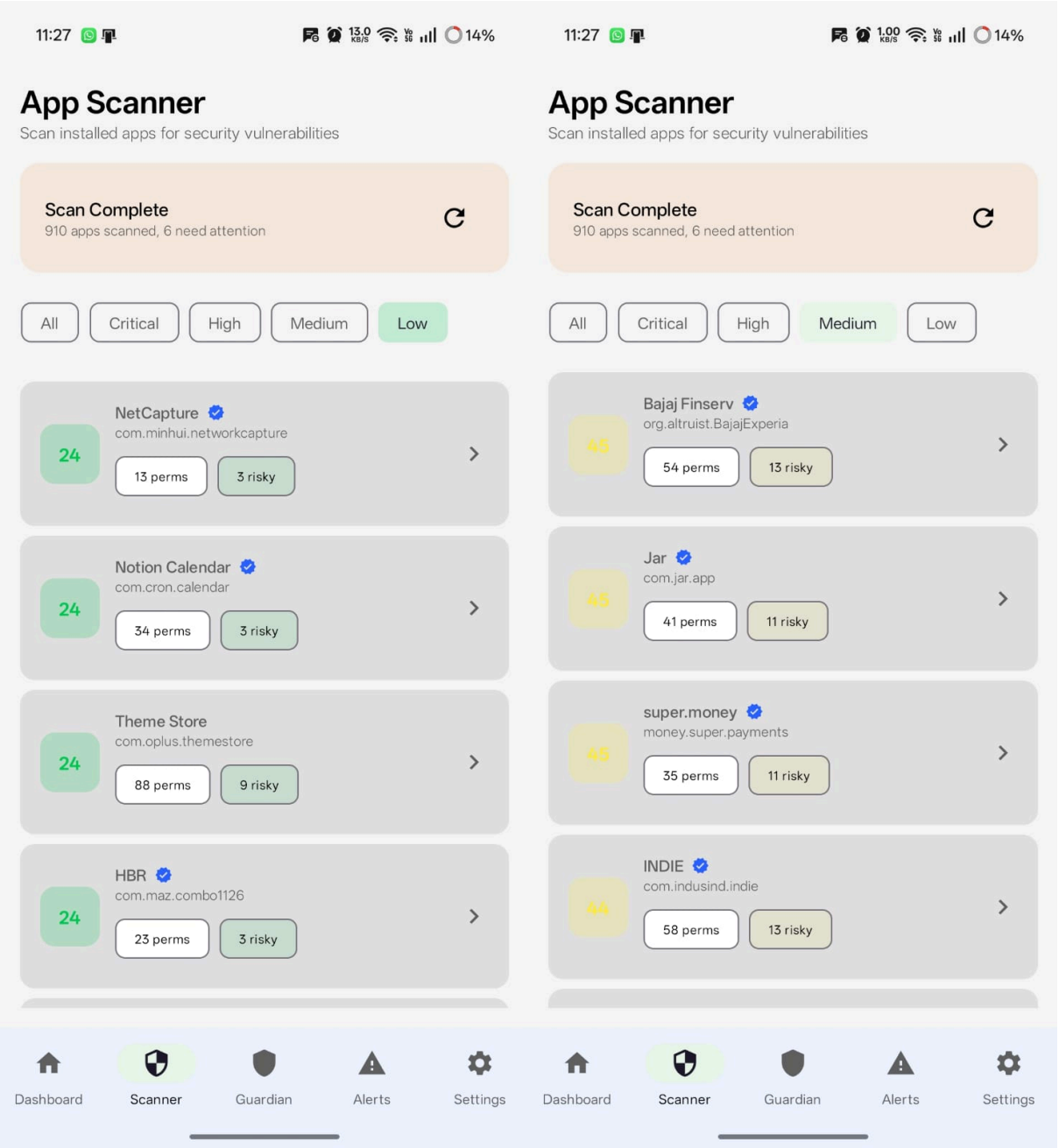
Feature	Innovation
	sensor *right now*.
Smart App Name Resolution (CLI)	Intelligent package-name-to-human-name parser with a 20+ entry cache for popular apps, handling edge cases like `com.google.android.apps.*` and `com.samsung.*` patterns.
Dark-Themed HTML Reports (CLI)	Generated reports use a modern dark UI with CSS grid, expandable cards, risk meters, and color-coded badges — not just raw text dumps.

11. Screenshots & Demo

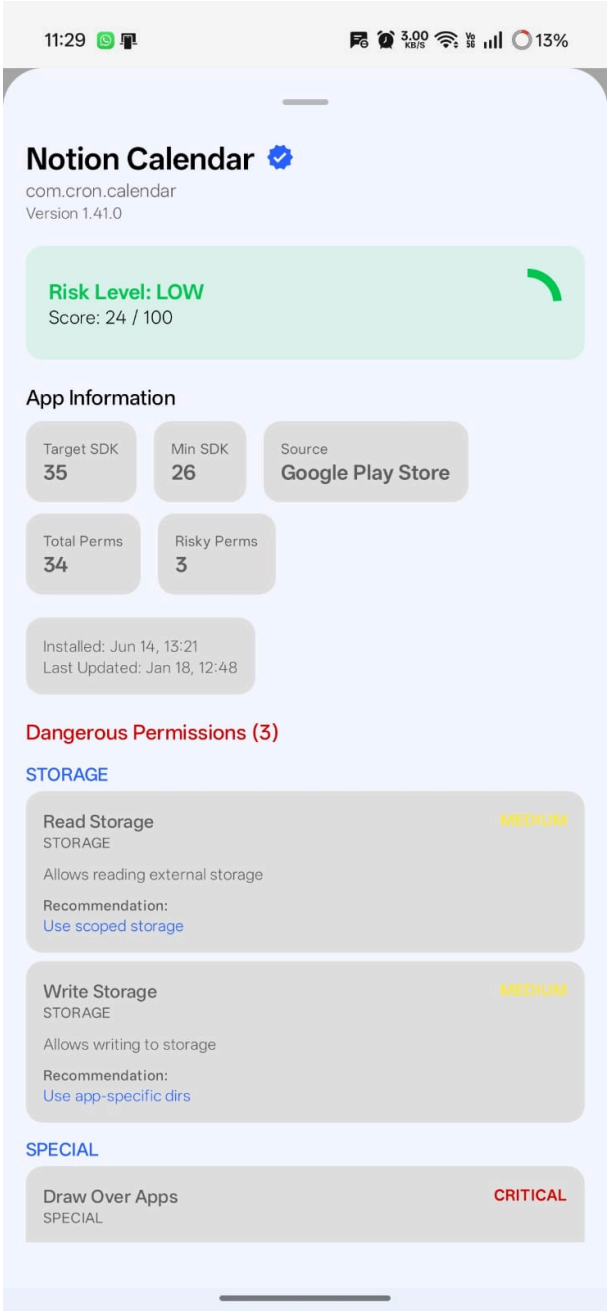
Android App Screenshots

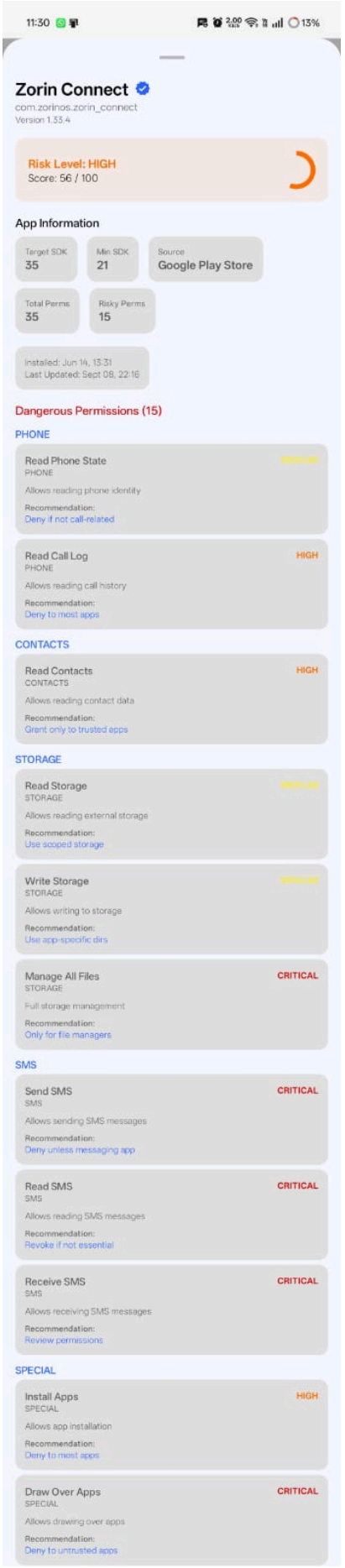
Screen	Description
Dashboard	Security overview with score gauge, stats, alerts, and sensor usage
Scanner — App List	Color-coded risk scores, Play Store verified badges, permission counts
Scanner — App Detail	Full security breakdown with permissions categorized and recommendations
Guardian	Real-time sensor toggles, active apps, 7-day usage trend chart
Alerts	Privacy alert cards with sensor filters and dismiss actions
Settings	Live permission status, Guardian config, data management, theme toggle

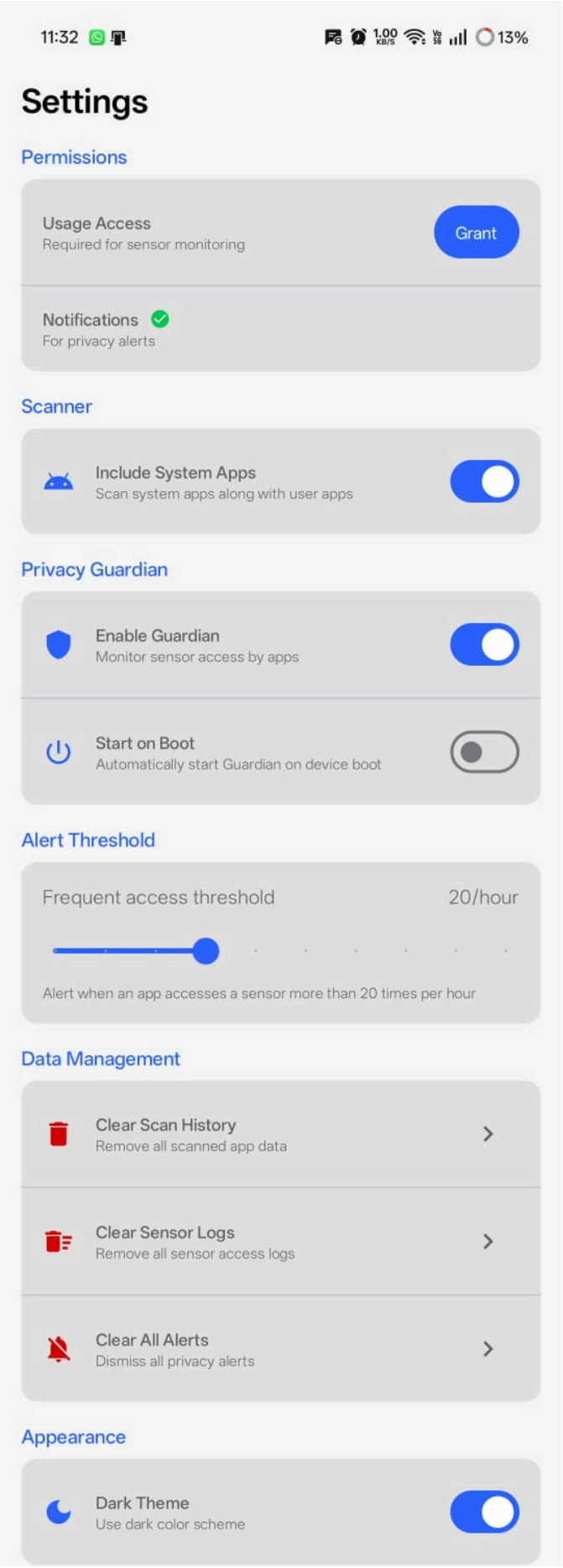

11:14
16%
Security Dashboard
Your device security overview
Security Score
Critical Risk
87
Quick Stats
910
Apps Scanned
6
High Risk
0
Alerts Today
High Risk Apps
View All
Clone Phone
16 dangerous permissions
HIGH
Score: 65
Zorin Connect
15 dangerous permissions
HIGH
Score: 56
Notes
2 dangerous permissions
HIGH
Score: 55
Recorder
HIGH
Dashboard
Scanner
Guardian
Alerts
Settings
11:25
14%
Clone Phone
16 dangerous permissions
HIGH
Score: 65
Zorin Connect
15 dangerous permissions
HIGH
Score: 56
Notes
2 dangerous permissions
HIGH
Score: 55
Recorder
2 dangerous permissions
HIGH
Score: 55
Translate
2 dangerous permissions
HIGH
Score: 55
Today's Sensor Usage
0
Camera
0
Mic
0
Location
0
Background
Quick Actions
Scan Now
Guardian
Dashboard
Scanner
Guardian
Alerts
Settings



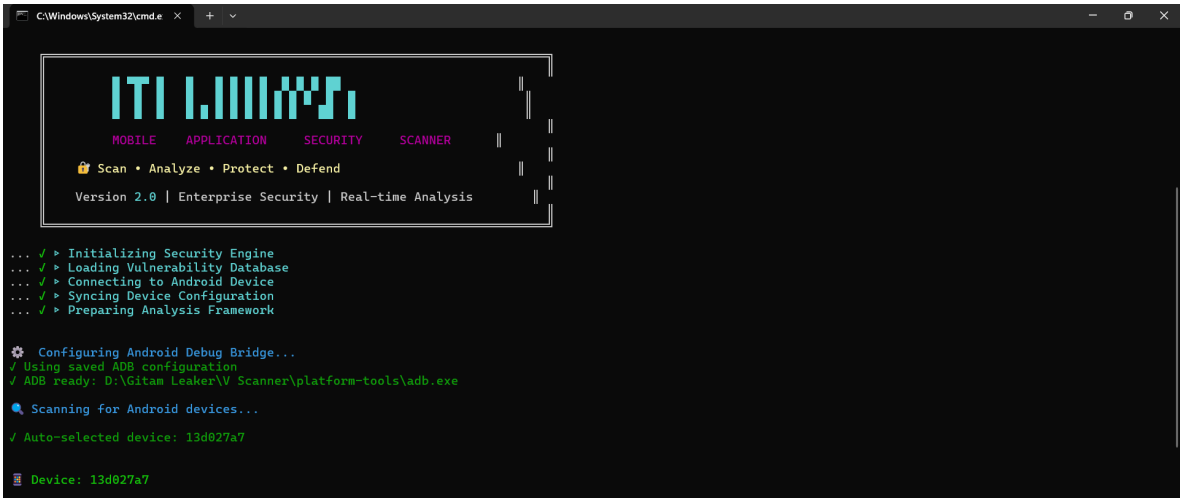
> **[SCREENSHOT PLACEHOLDER — App Detail Bottom Sheet]**
>
> *Expanded app detail showing: risk score gauge, installer source ("Google Play Store"), SDK info, total vs risky permission counts, dangerous permissions grouped by category, and the green "safe permissions" summary at the bottom.*

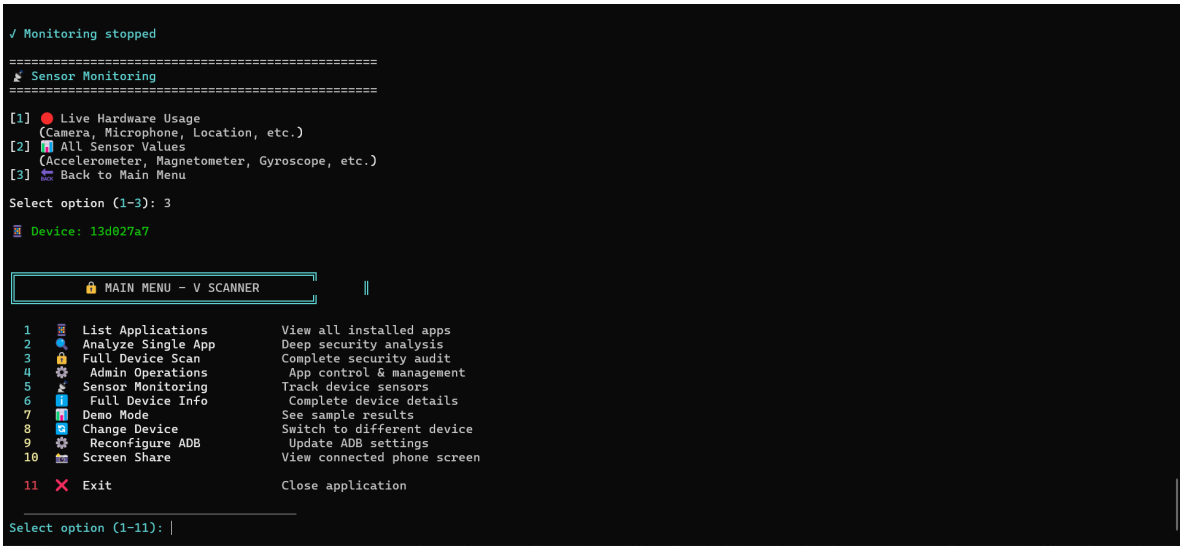

11:29
13%
Notion Calendar
com.cron.calendar
Version 1.41.0
Risk Level: LOW
Score: 24 / 100
App Information
Target SDK
35
Min SDK
26
Source
Google Play Store
Total Perms
34
Risky Perms
3
Installed: Jun 14, 13:21
Last Updated: Jan 18, 12:48
Dangerous Permissions (3)
STORAGE
Read Storage
STORAGE
MEDIUM
Allows reading external storage
Recommendation:
Use scoped storage
Write Storage
STORAGE
MEDIUM
Allows writing to storage
Recommendation:
Use app-specific dirs
SPECIAL
Draw Over Apps
SPECIAL
CRITICAL


11:30
13%
Zorin Connect
com.zorinos.zorin_connect
Version 1.33.4
Risk Level: HIGH
Score: 56 / 100
App Information
Target SDK
35
Min SDK
21
Source
Google Play Store
Total Perms
35
Risky Perms
15
Installed: Jun 14, 13:31
Last Updated: Sept 08, 22:16
Dangerous Permissions (15)
PHONE
Read Phone State
PHONE
Allows reading phone identity
Recommendation:
Deny if not call-related
Read Call Log
HIGH
PHONE
Allows reading call history
Recommendation:
Deny to most apps
CONTACTS
Read Contacts
HIGH
CONTACTS
Allows reading contact data
Recommendation:
Grant only to trusted apps
STORAGE
Read Storage
STORAGE
Allows reading external storage
Recommendation:
Use scoped storage
Write Storage
STORAGE
Allows writing to storage
Recommendation:
Use app-specific dirs
Manage All Files
CRITICAL
STORAGE
Full storage management
Recommendation:
Only for file managers
SMS
Send SMS
CRITICAL
SMS
Allows sending SMS messages
Recommendation:
Deny unless messaging app
Read SMS
CRITICAL
SMS
Allows reading SMS messages
Recommendation:
Revoke if not essential
Receive SMS
CRITICAL
SMS
Allows receiving SMS messages
Recommendation:
Review permissions
SPECIAL
Install Apps
HIGH
SPECIAL
Allows app installation
Recommendation:
Deny to most apps
Draw Over Apps
CRITICAL
SPECIAL
Allows drawing over apps
Recommendation:
Deny to untrusted apps


11:32
13%
Settings
Permissions
Usage Access
Required for sensor monitoring
Grant
Notifications
For privacy alerts
Scanner
Include System Apps
Scan system apps along with user apps
Privacy Guardian
Enable Guardian
Monitor sensor access by apps
Start on Boot
Automatically start Guardian on device boot
Alert Threshold
Frequent access threshold
20/hour
Alert when an app accesses a sensor more than 20 times per hour
Data Management
Clear Scan History
Remove all scanned app data
Clear Sensor Logs
Remove all sensor access logs
Clear All Alerts
Dismiss all privacy alerts
Appearance
Dark Theme
Use dark color scheme

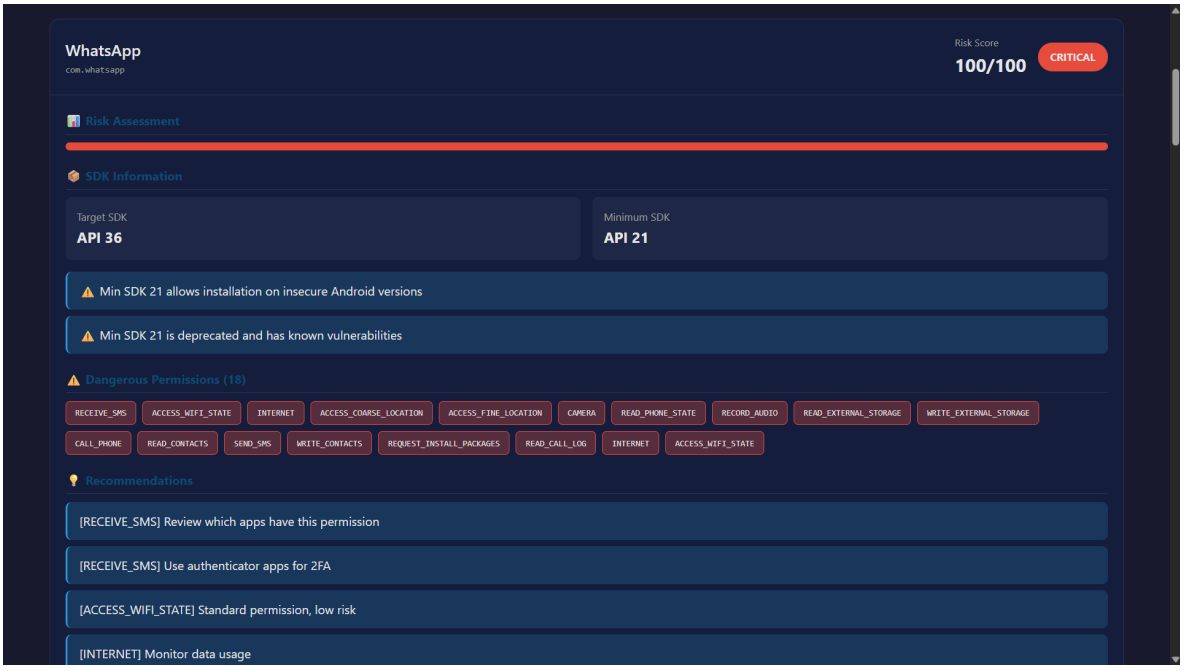
CLI Tool Screenshots





```
Select option (1-11): 2
Enter package name to analyze:  org.telegram.messenger

Enable deep APK analysis? (slower)
[1] No (quick analysis)
[2] Yes (search for URLs and hardcoded values)

Select (1 or 2): 1

🔍 Analyzing  org.telegram.messenger...

┌──────────────────────────────────────────────────────────────────────────────┐
│ 📊 Security Analysis: Messenger                                              │
└──────────────────────────────────────────────────────────────────────────────┘

Package:  org.telegram.messenger
Version: 12.4.3 (Code: 65272)
Target SDK: 35
Min SDK: 23

🔴 DANGEROUS PERMISSIONS (18):
  • Unknown - Allows approximate location access
  • Unknown - Allows precise GPS location access
  • Unknown - Allows initiating phone calls
  • Unknown - Allows reading call history
  • Unknown - Allows internet access
  • Unknown - Allows audio recording
  • Unknown - Allows viewing Wi-Fi connection info
  • Unknown - Allows reading external storage
  • Unknown - Allows reading contact data
  • Unknown - Allows modifying contact data
  • Unknown - Allows drawing over other apps
  • Unknown - Allows reading phone state and identity
  • Unknown - Allows access to device camera
  • Unknown - Allows requesting package installation
  • Unknown - Allows writing to external storage
  • Unknown - Allows location access in background
  • Unknown - Allows internet access
  • Unknown - Allows viewing Wi-Fi connection info

⚠️ SDK Issues:
  • Min SDK 23 allows installation on insecure Android versions

⚠️ RISK LEVEL: CRITICAL (Score: 100/100)

💡 Recommendations:
  • [ACCESS_COARSE_LOCATION] Prefer over fine location
  • [ACCESS_COARSE_LOCATION] Review necessity
  • [ACCESS_FINE_LOCATION] Use 'while using app' option
  • [ACCESS_FINE_LOCATION] Deny background location
  • [CALL_PHONE] Use dial intent instead
  • [CALL_PHONE] Review call logs
  • [READ_CALL_LOG] Deny to most apps
  • [READ_CALL_LOG] Clear call logs periodically
  • [INTERNET] Monitor data usage
  • [INTERNET] Use firewall apps
  • [RECORD_AUDIO] Grant only to call/voice apps
  • [RECORD_AUDIO] Revoke when not needed
  • [ACCESS_WIFI_STATE] Standard permission, low risk
  • [READ_EXTERNAL_STORAGE] Use scoped storage
  • [READ_EXTERNAL_STORAGE] Grant selective media access
  • [READ_CONTACTS] Grant only to trusted communication apps
  • [READ_CONTACTS] Review app reviews before granting
  • [WRITE_CONTACTS] Deny to most apps
  • [WRITE_CONTACTS] Backup contacts regularly
  • [SYSTEM_ALERT_WINDOW] Deny to untrusted apps
```

```
🔴 Live Hardware Usage Monitor
Real-time detection of active hardware access...
Last update: 23:23:15

┌──────────────────────────────────────────────────────────────────────────────┐
│ ⚠️ Unable to determine current app                                           │
└──────────────────────────────────────────────────────────────────────────────┘

📷 Camera Status
  🔴 ACTIVE Provider (android.hardware.camera.provider)
  🔴 ACTIVE Camera (com.android.camera)

🎤 Microphone Status
  🔴 ACTIVE Recorderservice (df24232com.android.soundrecorder.recorderservice)
  🔴 ACTIVE Soundrecorder (com.android.soundrecorder)

📍 Location (GPS) Status
  🔴 ACTIVE Google Play Services (com.google.android.gms)
  🔴 ACTIVE Nianticlabs (com.nianticlabs.pokemongo)

📊 Summary
  Camera apps: 2 active
  Mic apps: 2 active
  GPS apps: 2 active

⚠️ Apps are actively using hardware

[Press Ctrl+C to exit]

🔴 Live Hardware Usage Monitor
Real-time detection of active hardware access...
Last update: 23:23:19
```



HTML Report from CLI device full scan

security_report_20260303_230354.html

---

12. Installation & Setup

Android App

```
# Clone the repository
git clone https://github.com/ToTheBlankWorld/V-Scanner.git
cd V-Scanner/android

# Build the debug APK
./gradlew assembleDebug

# Install on connected device
adb install app/build/outputs/apk/debug/app-debug.apk
```

Or use the **pre-built APK**: `VScanner-debug.apk` (16.3 MB) available in the repository root.

CLI Tool

```
cd V-Scanner/cli

# Install Python dependencies
pip install -r requirements.txt

# Run the setup wizard
python setup.py

# Launch V-Scanner CLI
python main.py
```

Requirements

Component	Requirement
Android Device	Android 8.0+ (API 26+)
Android Build	JDK 17, Android SDK 35
CLI	Python 3.8+, ADB installed
Device Connection	USB debugging enabled OR wireless ADB

---

13. Project Structure

```
V-Scanner/
├── android/                          # Native Android Application
│   ├── app/
│   │   ├── build.gradle              # Dependencies & build config
│   │   ├── proguard-rules.pro        # R8/ProGuard rules
│   │   └── src/main/
│   │       ├── AndroidManifest.xml   # Permissions & components
│   │       ├── java/com/vsecurity/scanner/
│   │       │   ├── VScannerApplication.kt    # Hilt app entry
│   │       │   ├── scanner/
│   │       │   │   ├── AppScanner.kt         # Core scanning engine
│   │       │   │   └── PermissionDatabase.kt # 28 dangerous perms DB
│   │       │   ├── guardian/
│   │       │   │   ├── PrivacyGuardianService.kt  # Foreground monitor
│   │       │   │   └── BootReceiver.kt            # Boot persistence
│   │       │   ├── data/
│   │       │   │   ├── local/Database.kt          # Room DB v2 + DAOs
│   │       │   │   ├── model/AppModels.kt         # App entities
│   │       │   │   ├── model/GuardianModels.kt    # Sensor entities
│   │       │   │   ├── preferences/PreferencesManager.kt  # DataStore
│   │       │   │   └── repository/Repositories.kt # Data repos
│   │       │   ├── di/AppModule.kt               # Hilt DI module
│   │       │   └── ui/
│   │       │       ├── MainActivity.kt            # Single Activity
│   │       │       ├── navigation/Navigation.kt   # 5-tab bottom nav
│   │       │       ├── theme/Theme.kt             # Material3 theme
│   │       │       ├── screens/
│   │       │       │   ├── DashboardScreen.kt     # Security overview
│   │       │       │   ├── ScannerScreen.kt       # App scanner UI
│   │       │       │   ├── GuardianScreen.kt      # Sensor monitor
│   │       │       │   ├── AlertsScreen.kt        # Privacy alerts
│   │       │       │   └── SettingsScreen.kt      # Configuration
│   │       │       └── viewmodel/
│   │       │           ├── DashboardViewModel.kt
│   │       │           ├── ScannerViewModel.kt
│   │       │           ├── GuardianViewModel.kt
│   │       │           ├── AlertsViewModel.kt
```

```
│   │       │           └── SettingsViewModel.kt
│   │       └── res/
│   │           ├── drawable/          # Vector icons
│   │           ├── values/            # Colors, strings, themes
│   │           └── xml/               # Backup rules
│   ├── build.gradle                   # Root build config
│   ├── settings.gradle                # Project settings
│   └── gradle.properties              # JVM & build flags
│
├── cli/                              # Python CLI Tool
│   ├── main.py                       # Interactive menu (1532 lines)
│   ├── scanner.py                    # ADB interface (1876 lines)
│   ├── permissions.py                # Permission analyzer (679 lines)
│   ├── report_generator.py           # HTML/JSON reports (617 lines)
│   ├── ui_styles.py                  # Terminal UI (317 lines)
│   ├── setup.py                      # Cross-platform installer
│   ├── adb_setup.py                  # ADB path config
│   └── requirements.txt              # Python dependencies
│
├── docs/                             # Documentation
│   ├── ARCHITECTURE.md
│   ├── USAGE.md
│   └── ...
│
├── VScanner-debug.apk               # Pre-built debug APK (16.3 MB)
├── SUBMISSION.md                     # ← This document
└── README.md                         # Project README
```

---

14. Future Scope

Enhancement	Description	Impact
Network Traffic Analysis	Monitor real-time DNS queries and HTTP/HTTPS connections per app	Detect data exfiltration, tracking domains, C2 communication
ML-Based Anomaly Detection	Train a model on normal sensor access patterns, flag deviations	Reduce false positives, catch novel spyware behavior
Google Safe Browsing API	Cross-reference app-embedded URLs against Google's threat database	Identify known phishing/malware domains in APKs
VirusTotal Integration	Submit APK hashes to VirusTotal for multi-engine malware scanning	Leverage 70+ AV engines for known-malware detection
Permission Timeline	Track how an app's permissions change across updates	Detect permission creep (app gradually requesting more permissions)
Export & Share Reports	PDF/HTML export from the Android app + share-to-email	Enable organizational security audits
Multi-Device Dashboard	CLI tool aggregating security posture across multiple managed devices	Enterprise fleet management use case
iOS Support	Extend CLI to scan iOS devices via	Cross-platform organizational security

Enhancement	Description	Impact
	`libimobiledevice`	

15. Team & Repository

GitHub Repository

🔗 **https://github.com/ToTheBlankWorld/V-Scanner**

Quick Stats

Metric	Value
Total Source Files	35+
Lines of Code (CLI)	~5,000+
Lines of Code (Android)	~5,500+
Dangerous Permissions Tracked	28
Permission Categories	11
Trusted App Stores Detected	5
Room Database Tables	5
Preference Keys	16
Android UI Screens	5
CLI Menu Options	11
APK Size	16.3 MB

Built with ❤️ Team AVNI

V-Scanner — Because your apps shouldn't spy on you.